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Cloud computing adoption in Ghana; accounting for institutional factors

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ABSTRACT

In spite of increasing adoption of Cloud Computing (CC) by organizations in developed countries, the rate of adoption of the technology in sub-Saharan Africa has stagnated, although it has the potential to accelerate the speed of digital transformation. This paper contributes to the extant literature in this light by examining how the institutional environment influences the adoption of cloud computing. We investigated the role of mimetic, coercive and normative institutional pressures in predicting adoption outcomes of cloud computing among organizations in the region. After testing three main hypotheses and ten corollaries, and analyzing data collected from seventy-nine organizations with the partial least square structural equation modelling (PLS-SEM), the study found that the institutional pressures (mimetic, coercive and normative) explain 27% of the variance in cloud computing adoption. The mimetic – CC adoption path coefficient of 0.35 made the biggest contribution to the model while the normative – CC adoption path was the least contributor with a path coefficient of 0.18. The findings explain the determinants of adoption of CC in environments of low adoption and institutional challenges.

1. Introduction

Technological advancement brought about by development of new Information and Communication Technology (ICT) systems has accelerated productivity across the world. However, the huge costs involved in procuring and maintaining such large ICT innovation limit their implementation in many developing countries, particularly in Sub-Saharan Africa (SSA). Recent advancements in ICTs, especially in the development of cloud computing is helping organizations to overcome some of the constraints [1,2]. Cloud computing (CC) is the delivery of computing resources via the Internet as a service. The CC technology enables computing resources to be leased and released through the Internet in an on-demand basis. Its pay-per usage model, flexibility and scalability features has the potential to transform economic activities [3,4]. The value proposition of CC is radically different from the traditional on-premise computing deployment strategies as it enables the ownership and licensing burden, and support and maintenance requirements to be shifted to third party cloud service providers (CSP) [5,6]. Obviously, the scale economies and the modularity of use for economies constitute opportunities for ICT development in Africa.

Lee [6] and Askary and Kumar [2] portray CC as a new innovative technology with a lot of promise and Botta et al. [7] claim that it has not only become part of peoples' lives but also their future development. It is not surprising therefore that Yaokumah and Amponsah [8] aver that CC has become a suitable option for organizational development and performance. Consistent with Buyya et al.'s [9] argument that ICT has become a basic necessity, Joia and Marchisotti [10] suggest that CC helps to transform ICT into a basic utility service with an adequate infrastructure that enables users to access applications and data from any location. A report by Gartner [11] observed that "… CC would soon become the main strategy for many enterprises". This observation is manifest in the significant composition of CC in the technology portfolio of many contemporary organizations. In effect, CC innovation could contribute to the economic growth desired by developing countries. In its recent report, Gartner [12] reports that CC has now become the new norm as it evolves from a market disruptor to the expected approach for ICT.

Overall, CC adoption is growing around the world, even as the Infrastructure as a Service (IaaS), Platform as a Service (PaaS), and Software as a Service (SaaS) grew, and the services offered by Amazon Web Services (AWS) continue to be steady in recent times. The global CC market, which currently stands at about $350 billion, is expected to increase to about $650 billion by 2025, by which time 80% of all organizations are predicted to migrate toward the cloud [13], and

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serverless architecture, whose adoption rate is **26%** [14]. With a market share of 50% in 2018, serverless architecture was the fastest growing extended cloud service. The Flexera's report shows that 50% of enterprises spend more than $1.2 million a year for public cloud service, while another 13% spend the astonishing $12 million for the same amount of time. Although CC has become a foundation of digital business or silver bullet as Okai et al. [15] describe it, many organizations still struggle to optimize this powerful tool. The question is why? This question drives the study.

In spite of the increasing adoption rate of CC by organizations in developed countries, Dahiru et al. [4] argue that unavailability of broadband has contributed to the poor rate of adoption of the technology in SSA. Such argumentation may be moot in the presence of high international access to fiber optic in a country like Ghana where the price for an STM-1 bandwidth (155 Mbps) on submarine cables has fallen from $90,000/month to less than $15,000/month. It is therefore important to understand how the various players in the CC eco-system are interacting to take advantage of the opportunities offered by the technology. Beyond infrastructure and government policy issues [16], institutional challenges have been mentioned as central reasons for the low adoption of CC [17]. However, not much attention has been given to this in the empirical literature although institutional legacies are part of the historical origins in Africa which has impacted on the receptivity to new ideas and organizational forms (Ayentimi et al., [18]. This paper contributes to the extant literature in this light. The purpose of the study therefore is to examine how the institutional environment influences the adoption of CC. There have been some studies that have been done but approached the topic from the rational choice perspective based on frameworks like the TOE and DOI [19]. The rational approach suggests that developing countries like Ghana should be excited about the adoption of CC but the low adoption rates in the SSA region therefore limits its explanatory power. Indeed, many studies show that aside technological factors, the social, cultural and structural factors are key determinants of CC. Accordingly, the study employs the institutional approach to examine the cloud adoption rates in Ghana [3,10,20,21]. The study is situated in the literature that shows that context matters in explaining CC adoption [22]: [23].

Ghana is a lower middle income country, although such broad classification of countries tend to obscure the developmental differences among countries [24]. For instance, India, a middle income country has a number of local data centre operators which minimizes the costs of cloud computing, and reduces the impact of regulatory and legal issues of locating data centres outside the jurisdiction of data owners [25,26]. Bhat [26] also noted the existence of strong government support for CC in India, which may not necessarily be the case in a sub-Saharan African country like Ghana. Because of such nuances in technological and the environmental character, lessons from one country cannot be easily imported. The organizations that were involved in this study are members of the Association of Ghana Industries (AGI), and are part of the AGI's top 200 organizations, classified as large and medium businesses and are more likely to adopt CC technology. These organizations are also firmly established in the Ghanaian culture and, as such are good candidates for analyzing the cultural issues in adoption of CC. They include mining, energy, waste management, retail services and construction companies. This study is significant to the extent that it will deepen the understanding of determinants of adoption of CC in environments of low adoption [27]. As many countries in SSA are pursuing ICT for development, the outcome of this study will enrich current literature on organizational adoption of CC, and provide guidance for crafting policies for attaining ICT for sustainable development goals (SDG). The findings of this study could also serve as an important tool for regulators and policy makers who want to encourage the application of ICT in businesses to enhance productivity.

In the sections that follow, the literature reviewed is presented and thereafter the research methodology employed to achieve the research objective is described. The next section is the presentation of the study's findings and discussions after which the implications and concluding remarks are given.

2. Literature review

2.1. Overview of institutional theory

Meyer and Rowan [28] refer to an institution as a social structure made up of a collection of individuals or organizations within which collectives exercise action in a constrained environment. Collectives in this case or collectivism is a value that is characterized by emphasis on cohesiveness among individuals and prioritization of group interest over the self [29,30]. More recently, Reay et al. [31] have defined institutions as the taken-for-granted structures, practices, and meanings that define what people and organizations within their jurisdiction think, do, and aspire to. Raza et al. [32] also suggest that institutions originate from multiple rule-setting centres like governments, associations, and communities, all of which have an impact on organizational behavior and performance. Accordingly, institutional theory deals with the deeper and more resilient aspects of organizations and social structures. The theory also explain the processes by which certain structures and schemas, rules, norms, and routines, become established and accepted as authoritative guidelines for social behavior, by delving into the creations of such elements, their adoption, adaptation and diffusion over space and time; and how they fall into decline and disuse. In sum, the institutional theory posits that structural and behavioral changes in organizations are determined less by competition and the desire for efficiency and more by a need of organizational legitimacy [33–35]. This is consistent with Scott's [36] observation that institutions are comprised of regulative, normative and cultural cognitive elements that, together with associated activities and resources provide stability and meaning to social life" (pp. 48).

2.2. Institutional theory in Information Systems research

Institutional theory allows technology to be explored beyond the rational elements of designs of the technology and their economic impact, to include broader issues pertaining to the social, cultural and political aspects of technology [37]. King et al. [38] argue that institutional forces (regulatory forces) always have an inevitable presence in any attempt to understand information technology innovation across firms. They posit that institutions are influential forces with greater stability than the innovations they influence, but the methods of influence of institutions do change with time.

Studies that applied institutional theory have been conducted through different institutional theoretical lenses like DiMaggio & Powell's [35] isomorphic forces [33,39], Orlikowski's [40] Structuration theory of technology [41] and King et al.'s [38] Institutional intervention model [42]. According to Currie [43]; fewer studies in the IS field adopt the process method. When used in the IS field, institutions have been mostly regarded as an entity with effects and has been operationalized as the independent variable.

In their review of literature on Institutional Theory, Mignerat and Rivard [44] noted that even though Information Systems (IS) researchers acknowledge the existence of different units of analysis in institutional studies, most studies focus on the organizational level of analysis and neglect important units like the organizational field. Similar concerns were highlighted by Currie and Swanson [45] and Currie [43] who stressed the need to place the information technology (IT) artefact in their social, cultural and political contexts and criticized the neglect of the wider environment and individual units. This study conceptualizes technology (Cloud computing adoption) as the dependent variable, resulting from independent variables drawn from institutional influences that include technology (fashion perspective of cloud computing).

2.3. Institutional pressures

The strength of institutional theory is in its ability to account for the irrational elements in not only the structuration of organizations, but also the organization field. DiMaggio and Powell [35] attribute similarities in organizations to external forces acting on the organizations to form homogenous structures and cultures. The process that creates the homogenous structures in an organizational field, is what they refer to as isomorphism. The sources of these powerful forces were identified as the state, competition, and professions. Organizational actors make decisions they deem rational, however, isomorphic forces in the environment structure the range of options available to these actors (Berrone et al., 2013). DiMaggio & Powell [35] identified coercive, mimetic and normative isomorphism.

The next section provides details on how these institutional mechanisms emanates.

2.4. Research model and hypotheses

2.4.1. Mimetic pressure

Mimetic pressures cause an organization to change over time to become more like other organizations in its environment. According to DiMaggio & Powell [35] and Latif et al. [46]; mimetic pressure emerges when organizations are in competition to achieve superior performance. In an environment characterized by uncertainties and poor understanding of emerging technologies, ambiguous and lack of clarity, social actors (organizations) usually imitate structurally equivalent organizations deemed to be successful [35,47,48]. Although CC was once predicted to increase in developing countries like Ghana [49], uptake has been slow, and some studies have identified a dearth in knowledge and expertise as accounting for the low adoption rate [50]. According to DiMaggio & Powell [35], uncertainty and poor understanding of technology precipitate mimetic behavior. Yigitbasioglu [51] examined the role of top management team's (TMT) impact on the cloud computing for 1170 Australian firms, and found that mimetic pressures influence TMT's beliefs in the benefits of cloud computing. The results also show that TMT's beliefs drive their participation, which in turn affects the intention to adopt cloud computing solutions. Also, governments have been recognized as a key source of coercive pressure on organizations ([52,53]. We therefore, hypothesize that:

H1a. Mimetic pressure has a positive association with CC adoption.

H1b. Extent of CC adoption among governmental organizations has a positive association with CC adoption.

Trust is known to be shaped by cultural norms as noted by Hofstede [54] and also mediate the risk of putting third parties in charge of information resources [55]. Individuals or societies that subscribe to such collectivist worldview tend to find common values and goals as particularly salient, and demonstrate greater orientation toward a group ideology. Organizations will also copy organizations that are perceived to be successful [35]. Teo et al. [39] found this to be true for adoption of financial EDI. Accordingly, it is hypothesized that:

H1c. Extent of CC adoption among organizations known to potential adopters has a positive association with CC adoption.

H1d. The perception that organizations that have adopted CC have benefited from it has a positive association with CC adoption.

2.4.2. Coercive pressure

An organization experiences coercive pressures from other organizations it depends on. These pressures may be formal or informal, or they may arise from the cultural context of the organization [35]. DiMaggio & Powell [35] identified the government, parent organizations, and resource dominant organizations as sources of such coercive pressures. In SSA and other developing countries, coercive pressure usually emanates from the imposition of restrictive legal and regulatory conditions by foreign investors, key international suppliers and customers like the European Union, professional associations and multinational institutions like the World Bank (Latif et al., [46]. Empirically, Oredo et al. [3] employed the partial least square structural equation modelling (PLS–SEM) estimation to analyze the effect of institutional pressures on 93 Kenyan firms from the manufacturing, finance, and ICT sectors. The results indicate that coercive and normative pressures have a significant positive relationship with CC adoption but mimetic pressures did not have a significant effect. Likewise, Hsia et al. [20] investigate the drivers of e-health adoption and find that only mimetic and coercive pressures have a positive effect. Therefore we hypothesize that:

H2. Coercive Pressure has a positive association with adoption of CC

Montealegre [42] demonstrated that coercive pressure from governments are contextual and may promote or inhibit technology adoption. Other studies have also shown that government support influences E-Commerce adoption in SSA and other developing nations [56–58]. Suppliers of Cloud services also depend on wide distribution networks to get to the final customer. These agents of OEMs and Independent IT suppliers promote CC services in the local markets through supply side activities that position CC as the next legitimate infrastructure.

We therefore, propose the following hypotheses:

H2a. Government's promotion of CC is positively associated with CC adoption.

H2b. Regulatory barriers are negatively associated with CC adoption

H2c. Supplier promotion is positively associated with CC adoption

2.4.3. Normative pressure

Normative pressure from professional networks, universities, and other groups, are known to drive isomorphism [35]. Swanson & Ramiller [59] argued that the discourse involving technology vendors, consultants, potential users and academics among others, facilitate the process of interpretation of a new technology among a community of users. Lin et al. [21] examine the impact of institutional pressures and organizational innovation on e-business of 437 agricultural firms in China and report that normative pressure has a significantly negative impact on perceived barriers, while mimetic pressure and coercive pressure have no significant effect on perceived barriers. Similarly, Martínez-Ferrero and García-Sánchez [60] use a sample composed of 696 international non-financial firms from 16 different countries for the period 2007–2014 to show that the normative factor is the one that exerts the greatest explanatory power in the assurance demand, followed by coercive pressure, and mimetic was the least. We expect that in a collectivism society, the pressure to defer to group norms are even stronger, leading to the following hypotheses.

H3a. Normative pressure has a positive association with adoption of CC.

H3b. Participating in associations and networks that discuss CC has a positive association with adoption of CC:

IT is an institution with its own momentum that etches in the subconscious a taken-for-granted notion of validity [61]. Some heavily promoted IT innovations end up as fads or fashion [62]. A newer technology like CC that seeks to disrupt a more institutional alternative (Traditional Datacenter) will bear the suspicion of being a fad that may be adopted even though it is of spurious value, or may get rejected even though it offers real solutions to problems [63]. The rational choice between CC and private data centre, is therefore, structured by two separate institutional elements – the fashion proposition by CC and the cultural persistence of the private data centre [61]. Additionally, adoption increases when the degree of innovation is compatible or consistent with exiting values and needs of potential adopters [64]. Rogers also argues that innovations may be adopted because they are

perceived as a status symbol or enhances social recognition. Form the above arguments, it is expected that:

H3c. The perception that CC is an IT fashion is negatively associated with CC adoption.

H3d. The perceived lack of compatibility of CC is negatively associated with CC adoption.

H3e. The perception that CC is a status symbol is positively associated with CC adoption.

3. Research methodology

This study employed survey research (based on online questionnaire) to explain the relationship between institutional forces and CC adoption in Ghana. Two hundred Firms were purposively selected from the Association of Ghana Industries (AGI) database. About 1200 companies across all sectors are listed in the database but 200 classified as large and medium businesses were adopted as the sample frame for this study because they are more likely to adopt CC technology. One hundred and sixty-five (165) companies were contacted but sixteen (16) e-mail messages were undeliverable. One hundred and forty-nine (149) questionnaires were therefore considered to have been delivered. Seventy-nine (79) were returned for a response rate of 53%.

Respondents were encouraged to consult with other members of the organization if required, to capture the organization's position and the Structural Equation Modelling (SEM) technique was used in the analysis.

3.1. Measurement of variables

The questionnaire measured key variables for the study, which included CC adoption, institutional factors (mimetic, coercive, and normative) and key characteristics of the organizations studied.

3.1.1. Mimetic pressure

Following Teo et al. [39]; mimetic pressure was conceptualized as a second order formative construct; but emerging from three reflective sub-constructs, namely: Organizational Adoption (OrgAdop), Adoption by governmental organizations (GovAdop) and Perceived benefit for adopting organizations (PercBen). Adoption among organizations and government organizations were measured as perceptions rather than actual knowledge of adoption among entities.

3.1.2. Coercive pressure

Coercive pressure was operationalized as a formative construct, formed by pressure from suppliers and government. Unlike the Electronic data Interchange (EDI) context studied by Teo et al. [39]; CC adoption is unlikely to be affected by customer pressure. In line with Gibbs and Kraemer [53]; governmental pressure was conceptualized as government promotion and legal barrier. However, for practical purposes of defining reflective indicators for the sub-constructs, legal barrier was decomposed into two sub-constructs – legal prohibition and inadequate legal protection. Therefore, there were four sub-constructs for coercive pressure, namely; supplier promotion (SupProm), government promotion (GovProm), legal prohibition (LegProh), and Inadequate legal protection (LegBar).

Supplier promotion was measured by asking respondents the number of times in the last 12 months that they have been approached by suppliers. Government promotion was measured by two indicators: extent to which procurement laws supported CC and the extent to which government provides incentives for cloud usage. Inadequate legal protection (LegBar) was measured by perceptions of legal protection for data stored on third party servers and Legal prohibition (LegProh) was measured by perceptions about the extent to which business laws prohibit storage of data on third party servers.

3.1.3. Normative pressure

Normative pressure was conceptualized as a formative construct emerging from four reflective sub-constructs [39] namely: pressure from business and professional associations (Assoc), perception that CC is incompatible with the cultural environment (EnvIncom), perception that CC is a fashion promoted by interest groups (FashCC) and the perception that CC is a status symbol (ImaStat). Respondents were asked to respond "yes" or "no" to the questions; whether they belong to associations that have discussed or asked to adopt either CC or IT in the last 24 months.

The indicators were measured on five-point Likert scale based on Abrahamson [63]; 1996) and Moore and Benbasat [65] but the LegProh and whether they belonged to associations that used or asked to adopt CC were measured on Yes or NO scale.

3.2. Cloud computing adoption

The dependent variable CC adoption (CC-Adop) was conceptualized as reflective construct measured by three observable indicators adapted from Ref. [39]. Respondents were asked to indicate "yes" or "no" to questions on whether they have adopted cloud computing, whether they are considering adoption and whether they are likely to adopt in 12 months.

4. Results

4.1. Characteristics of respondents

As advised by Klein et al. [66]; the level of measurement and analysis were both congruent with theory at the organizational level as shown in Table 1.

About 80% of the organizations surveyed have been operating for more than 10 years with staff strengths of 50 or more. Most of the organizations can be considered to be fairly large in the Ghanaian context reflecting the sample frame made up of AGI. These organizations are well embedded in the Ghanaian culture and therefore serve as a good window to observe particularly the cultural influences on the structural properties of technology acquisition, particularly relating to CC. The organizations represented are from the mining, building and construction, retail services, and financial sectors (Table 2).

Table 3 shows the mean and standard deviations of the 28 indicators used in the study.

4.2. Validation of the measurement model (instrument validation)

4.2.1. Reflective indicator validation

The measurement model was first evaluated for convergent validity which attests to the fact that the indicators were understood the same way by respondents. This measurement is assessed by two factors. First, the indicator loadings must all have individual p values that are less or equal to 0.05, and second, each indicator loading must be greater than 0.5 [67,68]. Confirmatory factor analysis report in Table 4 shows that all indicators load well on their constructs with no cross loading problem, indicating that the indicators have convergent validity. Furthermore, the least indicator is 0.665 and are significant at $p < 0.001$. Average

Table 1
Positions of respondents.

Title	Number	Percentage
CEO	1	1.3
Director of IT/MIS	4	5.1
Head of Technology	13	16.5
IT Manager	35	44.3
Senior IT Specialist	12	15.2
Other	11	13.9
Non-Response	3	3.8

Table 2
Characteristics of participant organizations.

Organizational Characteristics	Category	Number (Frequency)	Percentage
Age of Organization	Less than 5 years	4	5.1
	5–10 years	11	13.9
	10–20 years	25	31.6
	20–50 years	22	27.8
	More than 50 years	16	20.3
	Missing Data	1	1.3
	Total	79	100
Industry	Financial Services	24	30.4
	IT	13	16.5
	Manufacturing	10	12.7
	Food and Beverage	3	3.8
	Oil and Gas	7	8.9
	Others	21	26.6
	Missing Data	1	1.3
	Total	79	100.0
Organizational staff strength	Less than 10	2	2.5
	10 to 20	2	2.5
	20–50	10	12.7
	50–100	10	12.7
	100–500	26	32.9
	More than 500	28	35.4
	Missing Data	1	1.3
	Total	79	100
IT Staff Strength	Less than 5	31	39.2
	5–10	18	22.8
	10–20	11	13.9
	20–50	15	19.0
	More than 50	3	3.8
	Missing Data	1	1.3
	Total	79	100

Table 3
Descriptive statistics of indicator variables.

Indicator Variables	Mean	Std. Deviation
Perceived Adoption Among Government	1.86	0.615
Perceived Adoption Among Organizations	2.34	0.597
Perceived Benefit to Adopters	3.86	0.593
Perceived Favorable Perception of Adopters	3.09	0.771
Government Incentive to adopters	2.00	0.751
Government Procurement Support for Adoption	2.58	0.942
Legal Protection Inadequate for using Third Party Servers	2.75	1.068
Legal Protection Inadequate for using foreign Third Party Servers	2.92	1.152
Legal prohibition to storage of data on foreign servers	4.61	0.608
Legal Prohibition of Storing Data on Third Party Servers	4.70	0.540
Supplier Pressure to Change to Cloud Services	2.27	1.106
Supplier Pressure to promote new CC services	2.33	1.047
Discussion of CC by Business Associations	1.36	0.483
Professionals Suggest CC adoption	1.62	0.488
Discussion of IT by Business Associations	1.59	0.494
Perceived Fashion Status of CC	4.37	0.803
Perception that CC will lead to loss of Jobs	3.82	0.874
CC is exaggerated	3.25	1.006
Software on PC is more Valued than Online Software	3.13	1.242
Organizations cannot run without traditional computing	3.23	1.213
Preference for TC because I can see servers and copies of software	3.41	0.968
Data is more valuable on your Computer than when stored in the cloud	3.53	1.096
Users of CC are more prestigious than nonusers	2.35	0.803
Users of CC are more Valuable than Non users	2.26	0.813
CC is a status symbol	2.36	0.953
Has Organization adopted CC	1.57	0.498
Is organization contemplating adoption of CC	1.71	0.459
Likely Adoption of CC in A Year	1.60	0.493

variance extracted shown in Table 5 are greater than the recommended threshold of 0.5 (Urbach & Ahlemann, 2010). Therefore the multiple indicators used in the study have convergent validity.

Discriminant validity was also used to attest to the fact that respondents did not associate questions for one construct with another construct (See Table 6). The terms in the diagonal (square root of Average Variance Extracted [AVE]) for each construct is higher than all correlations associated with the construct (terms in the same column above and below the latter), which shows that the constructs demonstrate discriminant validity [33,69].

To evaluate internal consistency as a measurement for reliability, composite reliability measures which is the preferred equivalent of Cronbach's alpha in SEM [70]; Urbach & Ahlemann, 2010) have been provided in Table 5. Composite validity measures must be greater than 0.7 [39,71]; Urbach & Ahlemann, 2010). The values for each construct meet that criterion; therefore the measurement model was internally consistent. Unidimensional reliability is assessable only in CB –SEM but assumed in PLS [72]. The reflective outer model is thus validated by the measurements reported above.

4.2.2. *Formative indicator validation*

Chin [73] advises that when second order latent variables are used, they must be validated. The first order sub-constructs become the indicators for the second order constructs. However, formative indicators have different validation scales from reflective constructs. Reliability measurements have no meaning in formative constructs, therefore they are not evaluated for reliability (Urbach & Ahlemann, 2010). Validity of formative indicators is however, assessed by examining the weights of the first order sub-constructs on the main constructs. They are validated if they are significant at a maximum p value of 0.05 (Urbach & Ahlemann, 2010). Another validity measure is the degree of multicollinearity which is determined by the variance inflation factor (VIF). The VIF is a measurement of indicator variance explainable by adjacent indicators of the construct. Unlike reflective indicators, there is little expectation of redundancy among formative indicators because they measure different dimensions of the same construct. As such, high VIFs undermine independence of formative constructs [74]. Stricter rules on multicollinearity call for a VIF threshold of 3.3 while more relaxed rules accept a threshold of 5 and sometimes 10 [71]; Urbach & Ahlemann, 2010).

Indicator weights from the study are reported in Table 7. It shows that all are significant at 0.05 level with the exception of one. However, the VIF values shown in Table 8 demonstrates strong independence among the formative LVs.

4.3. *Evaluating the theoretical model*

The structural model was tested using WarpPLS version 5. The results of the analysis are shown in Fig. 1 Table 9 and Table 10. From Fig. 1 and Table 9, the R^2 coefficient for CC adoption is 0.266 indicating that about 27% of variance in CC adoption is explained by the theoretical model.

Table 9 shows a summary of different measurements from the analysis. There are different schools of thought on the threshold for relevance of structural paths. While some authors ask for values greater than 0.02 [71], others suggest values of 0.1 or greater for a path to be considered to have meaningfully contributed to the model (Urbach & Ahlemann, 2010). As shown in.

Table 9, all three structural paths for this study are at least significant at 0.05 level with path coefficients higher than the more stringent requirement of 0.1. These values show that all three hypotheses H1, H2 and H3 have been confirmed. The corollaries to the main hypotheses were evaluated by studying the significance of the weights of the formative indicators shown in Table 7. All corollaries were supported apart from H3d, the perception that CC is a status symbol.

The mimetic – CC adoption path with path coefficient of 0.35 made the biggest contribution to the model while the normative – CC adoption

Table 4
Combined Loading and Cross Loading.

	AdopBen	GovProm	LegBar	LegProh	SupProm	Assoc	FashCC	EnvInco	ImaStat	CC-Adop	P value
AdopBen01	0.8	-0.058	-0.111	0.116	-0.177	0.021	-0.048	0.056	-0.125	0.237	<0.001
Adopben02	0.8	0.058	0.111	-0.116	0.177	-0.021	0.048	-0.056	0.125	-0.237	<0.001
GovProm01	0.075	0.879	0.004	-0.054	-0.025	0.048	0.004	-0.002	-0.051	-0.013	<0.001
GovProm02	-0.075	0.879	-0.004	0.054	0.025	-0.048	-0.004	0.002	0.051	0.013	<0.001
LegBar01	0.118	0.033	0.937	0.03	0.018	-0.037	0.009	0.019	-0.027	-0.067	<0.001
LegBar02	-0.118	-0.033	0.937	-0.03	-0.018	0.037	-0.009	-0.019	0.027	0.067	<0.001
LegPro01	-0.21	0.048	0.052	0.864	-0.08	-0.046	0.046	0.084	0.186	0.084	<0.001
LegPro02	0.21	-0.048	-0.052	0.864	0.08	0.046	-0.046	-0.084	-0.186	-0.084	<0.001
SupProm01	0.114	0.052	0.069	-0.051	0.945	-0.014	0.055	0.039	-0.005	-0.04	<0.001
SupProm02	-0.114	-0.052	-0.069	0.051	0.945	0.014	-0.055	-0.039	0.005	0.04	<0.001
Assoc01	0.056	0.044	0.091	0.023	-0.01	0.779	-0.151	-0.033	-0.047	-0.089	<0.001
Assoc02	-0.121	-0.175	0.038	-0.125	0.086	0.672	0.166	0.023	0.073	0.264	<0.001
Assoc03	0.045	0.1	-0.116	0.08	-0.06	0.834	0.008	0.013	-0.015	-0.13	<0.001
FashCC01	-0.216	-0.298	-0.111	0.004	-0.123	0.129	0.783	-0.068	0.113	0.16	<0.001
FashCC02	0.075	0.141	0.076	-0.144	0.422	-0.082	0.733	0.127	-0.013	-0.307	<0.001
FashCC03	0.164	0.187	0.045	0.147	-0.305	-0.059	0.698	-0.057	-0.113	0.143	<0.001
EnvIncom01	0.103	0.14	-0.027	-0.01	-0.052	-0.162	-0.012	0.797	0.022	0.064	<0.001
EnvIncom02	-0.239	0.213	0.011	-0.088	-0.212	0.106	-0.111	0.665	-0.261	0.119	<0.001
EnvIncom03	0.283	-0.285	0.228	0.162	0.04	0.133	0.07	0.727	-0.151	-0.187	<0.001
EnvIncom04	-0.17	-0.061	-0.198	-0.068	0.201	-0.05	0.042	0.762	0.349	0.008	<0.001
ImaStat01	0.135	0	-0.013	0.01	0.139	-0.067	-0.121	-0.022	0.836	-0.222	<0.001
ImaStat02	-0.053	0.078	0.04	-0.035	-0.079	0.056	0.011	-0.007	0.89	0.251	<0.001
ImaStat03	-0.079	-0.083	-0.03	0.028	-0.055	0.007	0.11	0.03	0.824	-0.046	<0.001
CC-Adop01	0.044	-0.09	0.026	-0.006	-0.128	-0.155	-0.02	-0.088	-0.027	0.765	<0.001
CC-Adop02	-0.049	0.121	0.089	-0.005	0.145	0.121	0.061	-0.142	0.086	0.807	<0.001
CC-Adop03	0.007	-0.033	-0.107	0.01	-0.022	0.024	-0.04	0.212	-0.056	0.86	<0.001

Table 5
Reliability measurements.

	AdopBen	GovProm	LegBar	LegProh	SupProm	Assoc	FashCC	EnvInco	ImaStat	CC-Adop
Composite Reliability	0.78	0.871	0.935	0.854	0.944	0.807	0.782	0.828	0.887	0.852
Average Variance Extracted	0.64	0.772	0.878	0.746	0.894	0.585	0.546	0.547	0.724	0.659

Table 6
Latent Variable Correlations and Square Root of AVE in the Diagonal.

	GovAdop	OrgAdop	AdopBen	GovProm	LegBar	LegProh	SupProm	Assoc	FashCC	EnvInco	ImaStat	CC-Adop
GovAdop	1											
OrgAdop	0.481	1										
AdopBen	0.367	0.434	0.8									
GovProm	0.494	0.387	0.299	0.879								
LegBar	0.062	0.254	0.084	-0.02	0.937							
LegProh	0.068	-0.024	0.021	0.058	-0.036	0.864						
SupProm	0.047	0.021	-0.038	0	0.014	0.075	0.945					
Assoc	0.256	0.199	0.051	0.243	0.114	-0.008	0.354	0.765				
FashCC	-0.061	-0.121	0.056	-0.003	0.041	-0.025	-0.074	-0.003	0.739			
EnvInco	0.093	-0.013	0.112	-0.033	0.212	0.108	0.046	0.083	0.277	0.739		
ImaStat	0.211	0.236	0.417	0.042	0.191	-0.141	-0.276	0.063	0.036	-0.02	0.851	0.052
CC-Adop	0.041	0.139	0.247	-0.011	0.227	0.078	0.409	0.286	0.044	0.123	0.052	0.812

Table 7
Indicator Weights.

	Mimetic	Coercive	Normative	P value
GovAdop	0.423			<0.001
OrgAdop	0.443			<0.001
AdopBen	0.405			<0.001
GovProm		0.424		<0.001
LegBar		-0.249		0.01
LegProh		0.67		<0.001
SupProm		0.465		<0.001
Assoc			0.187	0.042
FashCC			0.596	<0.001
EnvInco			0.617	<0.001
ImaStat			0.073	0.254

Table 8
VIF measures.

CC-Adop	Mimetic	Coercive	Normative
1.082	1.067	1.065	1.031

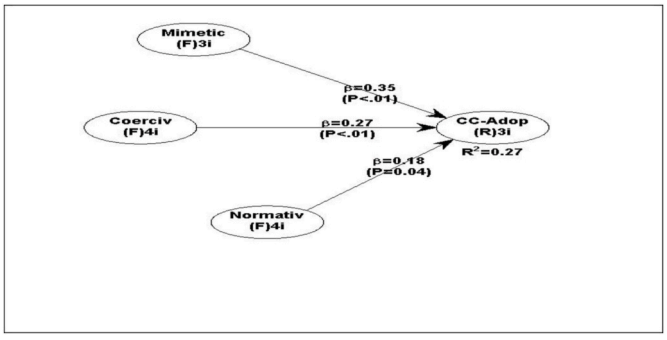


Fig. 1. Resultant structural model.

Table 9
Summary of analytic results.

Assessment Index	Mimetic	Coercive	Normative	CC-Adoption	
R-Squared coefficient	N/A	N/A	N/A	0.266	
Path Coefficients	0.35	0.272	0.183	N/A	CC-Adoption
Path Coefficient P values	<0.001	0.005	0.045	N/A	CC-Adoption
R-Squared contributions	0.137	0.091	0.038	N/A	CC-Adoption
Effective sizes for path coefficients	0.137	0.091	0.038	N/A	CC-Adoption

path was the least contributor with a path coefficient of 0.18. The resultant model after the analysis with path coefficients and weights of the sub-constructs is shown in Fig. 2.

Table 10 lists some important indices of the study. Among them is the Tenenhaus goodness of fit (GoF). The Q-square (Q^2) coefficient also known as the Stone-Geisser test is a measure of the predictive relevance of the model [70,75], hence may be viewed as a model validity test. The reported value of 0.28 (greater than zero) attests to the relevance of the model. The SPR value of 1 (greater than 0.7) gives an indication of a causality in the model.

Table 10
Model fit and quality indices.

	Value	Note
Average Path Coefficient	0.269	P = 0.003
Average R-Squared	0.266	P = 0.003
Average Full Collinearity	1.06	acceptable if ≤ 5, ideally ≤ 3.3
Tenenhaus GoF (GoF)	0.353	small ≥ 0.1, medium ≥ 0.25, large ≥ 0.36
Sympson's paradox ratio (SPR)	1	acceptable if ≥ 0.7, ideally = 1
R-squared contribution ratio (RSCR)	1	acceptable if ≥ 0.9, ideally = 1
Statistical suppression ratio (SSR)	1	acceptable if ≥ 0.7
Q-squared coefficient for CC Adoption	0.282	>0 for predictive validity

4.4. Discussion

The results of the study show that institutional factors influence organizational behavior towards adoption of CC in Ghana. Institutional determinants were confirmed to be operational through all the three pillars that frame institutions [35,36]. The study found support for three mimetic influences from the adoption of CC. The largest institutional influence was from mimetic pressure. Other organizations' adoption of CC and the perception that the adopters were successful encouraged others to adopt CC. Governments' role in institutionalization has mostly been captured as coercive, but was hypothesized to be a source of mimetic pressure because in SSA, governments are conspicuous in IT adoption. While previous studies conceptualized mimetic pressure as copying competitors [39,51], the findings of the study suggest that it is more of copying known organizations, especially those identified as successful. It is worth mentioning that Lin et al. [21] and Liu et al. (2010) show that mimetic pressure does not have an effect on perceived barriers to E-business and eSCM adoption respectively in China. Martínez-Ferrero and García-Sánchez [60]; however, show that mimetic factors have the least explanatory power in voluntary assurance demand for large global firms, and Hsia et al. [20] report that mimetic pressures have the greatest impact in e-health adoption. The difference could be due to the sectors or industries considered, size of firms and more critically the sociocultural, economic and political context in which the organizations operate.

Secondly, the perception about the kind of prohibitions placed on data transfer seems to be very important in shaping CC adoption. Coercive pressure has an impact on CC with the largest contributor attributed to legal prohibition. Government promotion activities and supplier promotion also had a positive influence on CC adoption. This finding emphasizes the contextual nature of technology adoption and supports the assertion that models of IT adoption may not be imported wholesale from one context to another [76]. Unlike Teo et al. [39] who found normative pressure to be the most significant institutional pressure in the context of electronic data interchange (EDI) adoption, this study found normative pressure to be the least significant contributor to the model. Indeed, Lin et al. [21] find that normative pressure has a significantly negative impact on perceived barriers, while mimetic pressure and coercive pressure have no significant effect on perceived barriers to e-business adoption. Finally, the findings of the study do not support the view that CC is a fashion promoted by interest groups and thus did not find support for the argument that CC is a status symbol.

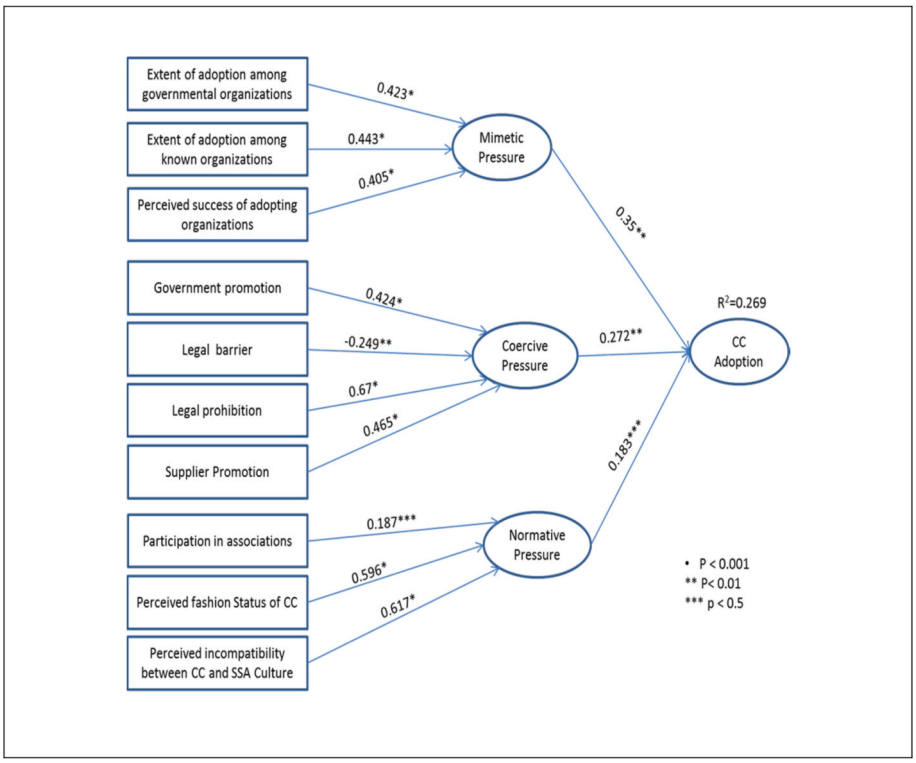


Fig. 2. Resultant framework after analysis.

5. Conclusion

This study examined the institutional determinants of CC adoption in Ghana through a theoretical model that was deduced through a review of previous studies and empirical evidence. From the analysis of 79 responses received from organizations, the results show that institutional forces influence CC adoption. Mimetic, coercive and normative forces explained 27% of the variance in CC adoption. These findings are consistent with Hsia et al.'s [20] argument that institutional forces enable adaptive changes in not only the competitive and regulatory environments but also the technological arena. Also, the findings of the study provide strategic impetus for CC adoption by organizations in their bid to be competitive by understanding the environment in which they operate. Obviously, the institutional pressures influence the application of ICT but more importantly CC adoption. The findings suggest a number of implications for researchers, policy makers and practitioners. First, to promote CC adoption, and improve organizational competitiveness, the critical role of institutions must be understood and strengthened. This calls for management of organizations and the government to drive technological advancement and build the necessary capacity to be able to exploit the opportunities that the new market environment presents.

Second, it is important that the IS practice community is aware of the potential of institutional forces in shaping the adoption and use of CC. The identified institutional pressures stimulate organizational norms and beliefs required to create an atmosphere that supports environmental sustainability. It is worth mentioning that the results of the study are limited as data was collected from the Ghanaian context and therefore does not necessarily apply to other African contexts. Additionally, this is a cross-sectional study and therefore future studies could employ longitudinal designs to improve the robustness of the results.

Author statement

Joseph Adjei: Conceptualization, Methodology, Writing – original draft, Visualization, Supervision, Analysis and Validation, Writing – Reviewing and Editing. Samuel Adams: Conceptualization, Methodology, Data curation, Visualization, Supervision, Analysis and Validation, Writing – Reviewing and Editing. Lovestone Mamattah: Conceptualization, Methodology, Data curation, Writing – original draft, Analysis and Validation, Writing – Reviewing and Editing.

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